Individual Activity Pattern Analysis Using Mobile Signal Data: A Case Study in Nanjing

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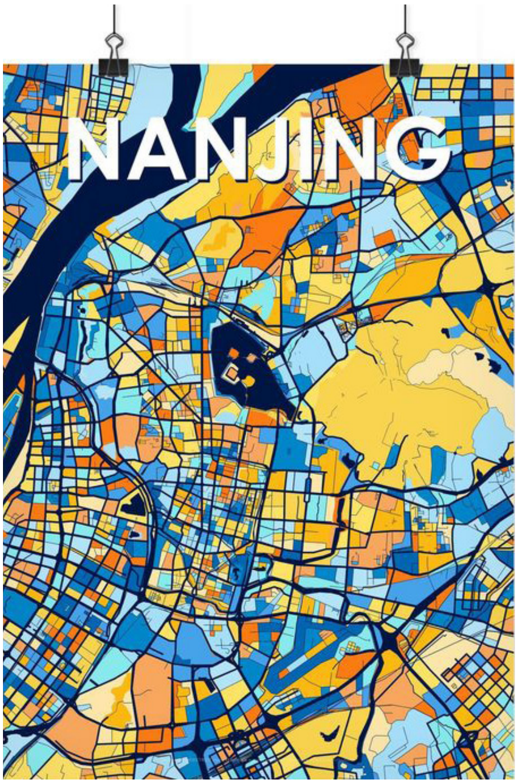
RESEARCH BACKGROUND&PURPOSE



The fascinating aspect of urban planning is that it often involves thinking at different scales. Sometimes, planners need to put themselves in the shoes of community members and work to secure equal housing and educational opportunities. Other times, planners need to take a bird's-eye view of the entire city and plan road networks to promote more balanced development, or observe to gain an understanding of the overall movements of city residents. For my studies in urban planning, classroom content often involves saving structurally declining areas of cities or urban renewal, which often focus on a specific neighborhood or individual case. However, to have a deeper understanding of a city as a whole, it is necessary to think and study at multiple scales. When faced with the task of controlling the entire city or larger scales, we often need to use the thinking mode and analytical tools of the corresponding scale, which often means combining quantitative thinking and more quantitative tools and means. That is why I have spent over two years participating in this research project. I have a very high interest in urban planning and its interdisciplinary fields and hope to gain a deeper understanding of city residents and the city itself by standing at a larger scale through this still in-progress research. I also hope to expand my quantitative analysis capabilities through this research.

Traditional urban transportation patterns are typically assessed through methods such as observation and surveys, combined with experience-based judgment. In contrast to the more extensive and inefficient traditional approaches, the collection of information on human activities and congregation patterns in the city through spatial big data allows for a more comprehensive observation and analysis of people's movement behaviors. Meanwhile, the scope of research has also been less constrained. With the increasing maturity of data collection technologies, such as mobile signaling data, Points of Interest (POI), social media data, and GPS positioning data, these data sources have started to be applied in research on urban vitality.

Currently, related studies and scholars have proposed various methods to describe and observe urban transportation patterns using available spatial big data and research methodologies. This study, however, primarily utilizes mobile signaling data from residents in Nanjing to explore human activity patterns in a simpler and more efficient way.

This research seeks to integrate and summarize human activity patterns at the city scale and aims to provide relevant research support for urban planning to some extent.

Given the long time span of this study, it is divided into two main phases.

PHASE I:

Research on Overall Urban Activity Patterns and Vitality in Nanjing

The first phase of the study focuses on data cleaning and preliminary analysis. Its aim is to gain an overall understanding of the behavior and activities of Nanjing residents based on mobile signal data, including the overall distribution of workplace and residential locations, the crowd aggregation distribution, and the general spatiotemporal distribution. This information helps in managing the overall data and lays the groundwork for Phase 2, especially for further research on smaller-scale patterns.

The mobile signal data used in this study includes the following information: user ID, time of mobile signal generation, and the spatial coordinates (longitude and latitude) of the mobile signal.

1.1 Workplace and Residential Area Identification Based on Mobile Signal Data

According to literature and the characteristics of the data, workplace and residential area analysis can be conducted in various ways. For example, analysis can be based on the proportion of different types of POIs (Points of Interest) in each block, or by analyzing the difference between outbound and inbound trips during peak hours, or a hybrid approach. This study uses the ArcGIS Pro platform to process the difference between outbound and inbound trip data during the early morning peak period (06:00-09:00). It assumes that areas where outbound trips exceed inbound trips are residential areas, while others are identified as workplace areas. To simplify the process and follow the logical approach, the study divides Nanjing into 50m x 50m grids, creating a rough distribution map of workplace and residential areas within the city (Figure 1).

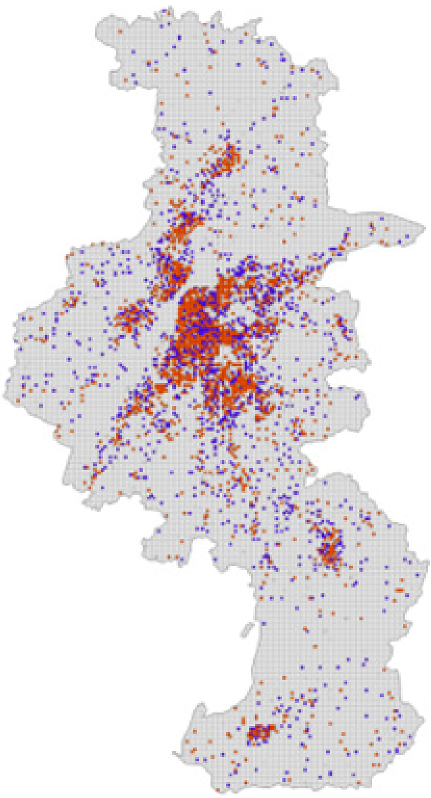


Figure 1
Workplace and Residential Area
Identification Result
(Residential Area:Orange Workplace: Blue)

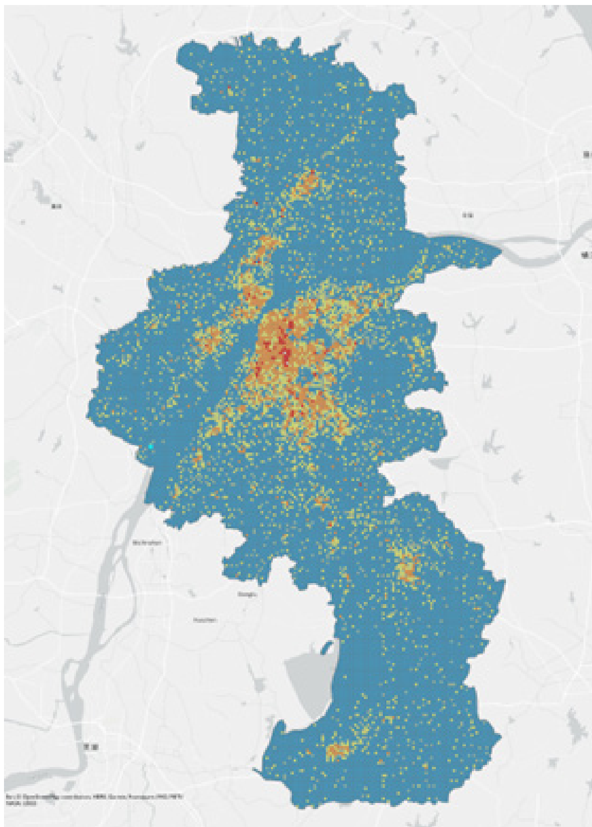


Figure 2
Crowd Aggregation Spatial Distribution

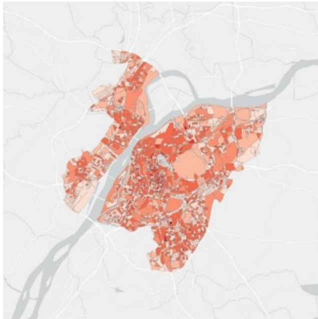
1.2 Analysis of Crowd Aggregation Spatial Distribution

The analysis of urban mobility in Nanjing is based on a grid-based approach, measuring the density of mobile signal occurrences to assess the city's traffic state. Data is extracted from 100,868 mobile signal records collected over a 24-hour period on November 4, 2019 (Monday) in Nanjing. The data is divided into 50m x 50m grids, as described earlier.
The mobile signal data is then connected with the ArcGIS fishnet grid and the count of mobile signal records within each grid is calculated. To eliminate the impact of irregular grid shapes near the boundaries of Nanjing's administrative area, only data within the grid's interior is used to calculate data density (Figure 2).

1.3 Analysis of Temporal and Spatial Characteristics of District-Level Mobility

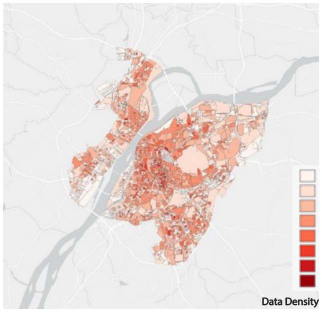
Districts are used as the unit of analysis, with the density of mobile signals indicating the urban transportation characteristics in Nanjing's central area during different time periods within a day. Data selected: 128,386 mobile signal records collected over a 24-hour period on November 4, 2019 (Monday). Based on multiple literature sources, a reasonable division of time is made, splitting the day into five time periods: (00:00-06:00, 06:00-09:00, 09:00-17:00, 17:00-20:00, 20:00-24:00).
Given the primary information in the existing data, time and spatial attributes will be the focus of the analysis. To continuously reflect the traffic mobility patterns in the city, performing spatiotemporal clustering on the mobile signal data can help to identify high-activity transportation areas over time.

Due to the varying amounts of mobile signal data across different time periods, the aim of this study is to conduct a horizontal comparison to identify high-traffic flow areas during different time segments. Therefore, during the visualization process, no unified classification standard for mobile signal density is applied. Instead, the data for each time period is classified based on geometric intervals.



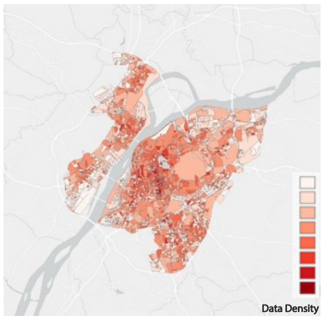
Time Segment (1): 00:00-05:59
Data Number:8041

Time Segment (1):
Although the window period is long, the amount of mobile signal data is relatively small, and the distribution is quite homogeneous. The city's traffic flow is low, with no significant aggregation patterns, and traffic generally disperses from workplace areas to residential areas.



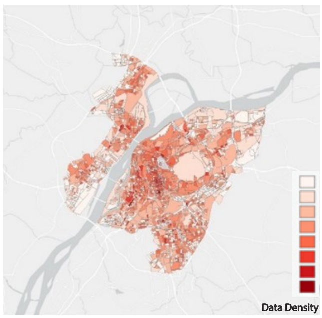
Time Segment (2): 06:00-08:59
Data Number:16640

Time Segment (2):
This is the early morning peak period, where the volume of data increases and a pattern of aggregation in certain districts starts to emerge. Compared to the homogeneous distribution in (1), segment (2) reflects the distribution of urban traffic flow during the early peak period to some extent.



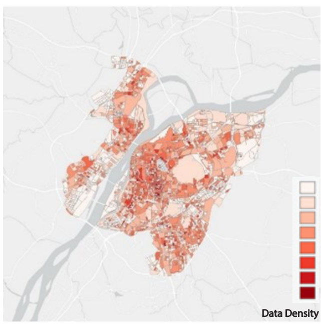
Time Segment (3): 09:00-16:59
Data Number:68567

Time Segment (1):
This time window is the longest, with the largest amount of data. During this period, high-traffic flow peaks occur in different land-use areas such as workplace and residential zones. As a result, the data shows partial aggregation but remains slightly dispersed overall.



Time Segment (4): 17:00-19:59
Data Number:24151

Time Segment (1):
Traffic flow in the main city and commercial districts continues to rise. Besides commuting between workplaces and residential areas, non-essential trips contribute to the increased data volume. Meanwhile, other non-core areas resemble the homogeneous distribution seen in (1), reflecting the dynamic of traffic dispersal from workplace areas to residential zones.



Time Segment (5): 20:00-23:59
Data Number:10987

Time Segment (1):
In some areas of the main city, data density remains relatively high, while other regions show lower data density. This suggests that the evening peak commute between workplaces and residential areas has already ended, and traffic flow in residential areas has decreased.

The above methods demonstrate the distribution of mobile signal data in different districts of Nanjing's central urban area across different time periods. While this approach can reflect the urban transportation state to some extent, the slice-based observation still fails to provide a complete and objective reflection of the city's transportation dynamics. Therefore, a continuous, comprehensive spatiotemporal analysis approach is needed.

1.3 Clustering Analysis

ST-DBSCAN is an extension of the DBSCAN algorithm in the time dimension, which assumes that the categories can be determined based on the density of sample distribution. ST-DBSCAN describes the density of sample distribution within a spatiotemporal neighborhood using the parameters (Eps1, Eps2, Minpts). The spatiotemporal neighborhood, Eps, is formed by the spatial neighborhood Eps1 and the temporal neighborhood Eps2. Minpts refers to the threshold number of samples required in the spatiotemporal neighborhood for a sample to be considered part of a cluster. ST-DBSCAN inherits the characteristics of the DBSCAN algorithm, including not requiring the number of clusters to be pre-defined, and being able to discover clusters of arbitrary shapes even in noisy datasets.

Parameter Selection:

- **Density Threshold (Minpts)**: This refers to the minimum number of spatiotemporal objects required within the spatiotemporal neighborhood. It is generally calculated as Minpts ≈ ln(N), where N (20,374) is the total number of spatiotemporal objects involved in the calculation. ln(20,374) ≈ 9.92 ≈ 10.
- **Spatial Radius (Eps1)**: This is determined through the K-Dist plot. From the plot, it can be seen that the distance before the K-th nearest spatiotemporal distance drops rapidly at 1000 meters, and then stabilizes thereafter. Therefore, the spatial radius is set to 1000 meters.
- **Temporal Radius (Eps2)**: This is generally determined based on the data source and the scale of analysis. It needs to ensure both the effectiveness and reliability of the results, while also considering practical circumstances and computational efficiency. It is typically determined by referring to the research direction and adjusting based on the experience of other scholars. After referring to other studies and adjusting, the temporal radius is set to 1200 seconds.

The data used in this study consists of 20,374 randomly sampled mobile signal records collected over 24 hours on November 4, 2019 (Monday). As shown in the figure, the data includes encrypted user IDs, date and time, longitude, and latitude. To reasonably process the existing data, it is necessary to effectively utilize the characteristics of the different attributes. The time and space coordinate information can be used for spatiotemporal clustering to obtain three-dimensional clusters containing both spatial and temporal information.

Under the parameter settings of (Eps1=1000, Eps2=1200, Minpts=10), clustering of 20,374 input data points resulted in 176 valid clusters, totaling 6,329 valid points and 14,045 discrete points.

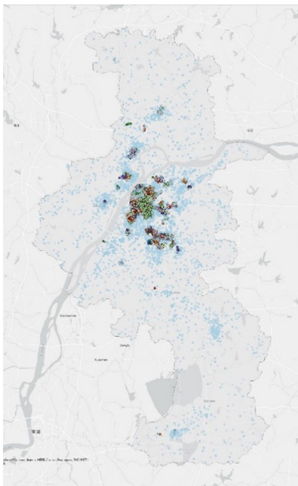


Figure 3
Distribution of point clusters in Nanjing (light blue indicates discrete points)

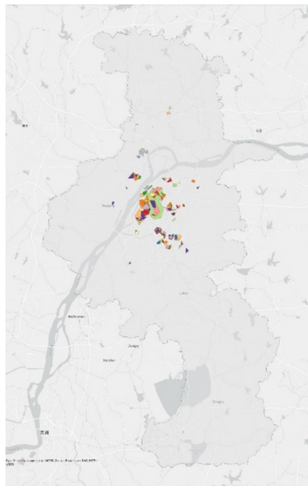


Figure 4
Distribution of the minimum spatial range enclosed by point clusters in Nanjing

Observing the distribution of valid point clusters on the plane, it can be seen that the clusters are primarily concentrated in the main urban area of Nanjing and in the core development zones of other regions. After transforming the data into three dimensions, the spatiotemporal regions where the clusters are located become more visually intuitive (Figure 5).

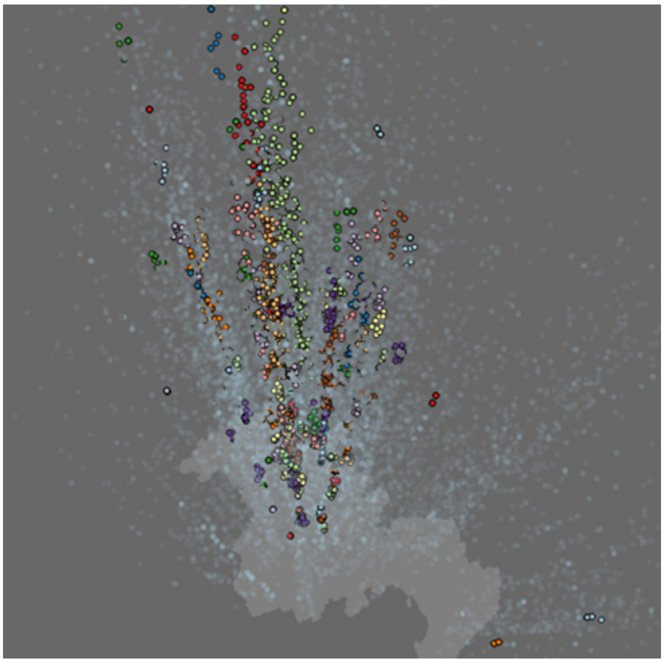


Figure 5
Visualization of Point Clusters in 3D Spatiotemporal Space in Nanjing

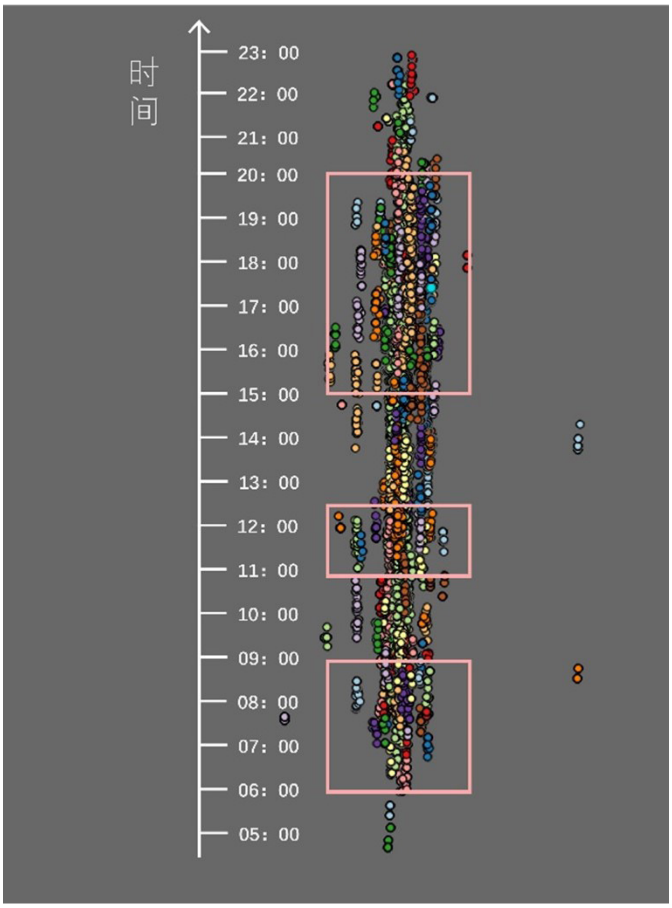


Figure 6
Distribution of Clustered Point Clusters in the Temporal Dimension

As shown in figure 6, during the three time windows of 06:00-09:00, 11:00-12:30, and 15:00-20:00, a large number of concentrated spatiotemporal clusters were formed. Some areas, due to denser mobile signaling data, generated several clusters that spanned a larger cross-domain spatiotemporal range, indicating large-scale high-traffic travel areas

This part of the study primarily uses the visualization of mobile signaling data to explore the distribution characteristics of population mobility in both time and space. By combining the spatial distribution of urban traffic infrastructure in Nanjing and trends in population movement, it analyzes the main factors causing the differences in spatiotemporal distribution. Furthermore, the ST-DBSCAN clustering method reflects the spatiotemporal aggregation and dispersion of the population within Nanjing. Analysis of low-vitality zones is conducted, identifying areas that require human-centered transformation.

The ST-DBSCAN clustering above reveals the spatiotemporal aggregation status of mobile signaling data within the city of Nanjing, reflecting the spatiotemporal status of traffic flow during working days to some extent.
Combined with the earlier research on the distribution of residential and workplace locations, it provides a macro perspective for human-centered urban transportation reform. This will ultimately help improve the quality of life for urban residents and reduce traffic and environmental pressures.

PHASE II:

Individual Activity Pattern Analysis

The research in the second phase differs from the first phase. While the first phase focuses on an overall perception of the data, the second phase is more centered on the study of individual activity pattern. The second phase is an extension of the first phase, aiming to cluster and analyze the activity patterns of Nanjing residents. Therefore, the key task in the second phase is to process mobile signal data and extract information that reveals activity patterns.

The overall approach in this phase is as follows: assign semantics to stop points and identify each stop point as a corresponding activity. After this, the individual's activities are linked together into activity chains, followed by a series of data processing, analysis, and clustering to finally summarize the individual activity patterns of different population groups.

Phase II requires more complex data processing and more frequent use of various Python algorithms. Since this research is still in the paper publication stage, only the research methodology is presented here. Some specific steps and code of the core algorithms are not shown.

2.1 Activity Chain Construction

Activity chain construction requires linking behavioral activities together, which means that the identification of behavioral activities must be done first. The key to identifying behavioral activities lies in analyzing the mobile signal data and assigning semantics to each data point through algorithms.

2.1.1 Identification of Work, Rest, and Other Activities

The first step is to roughly identify three main types of activities through mobile signal data: work, rest, and other activities. The main reason for this is that work and rest activities generally last for a longer duration and tend to have regularity and periodicity. Therefore, these two types of activities are prioritized for identification.

The data processing approach is as follows:

The most frequently visited stop points between 9:00 PM and 8:00 AM are marked as potential residence (H). The most frequently visited stop points on workdays from 9:00 AM to 5:00 PM are marked as potential workplaces (W).The remaining stop points are marked as potential other activities (O).

Considering that the study period includes an average of 19 workdays and 11 rest days per month, the stop points with more than 10 appearances and a total stay time of over 40 hours are designated as actual residences; stop points with more than 7 appearances and a total stay time of over 28 hours are designated as actual workplaces; the remaining stop points are classified as other activities.

2.2.2 Identification of Other Activity Types

In addition to work and rest, other activity types need to be further subdivided and identified to construct a more accurate and detailed activity chain. In this step, besides using mobile signal data from Nanjing, base station coordinate data and POI data from Nanjing for November 2019 are also used. Based on the characteristics of the data and research needs, the POI data will be filtered and reclassified into four categories. The other activity types identified in Section 2.1.1 will also be further categorized into four specific activities: shopping, culture, school, and medical.

At this stage, the POIs contained within the range of the Voronoi polygons defined by the base station coordinates are considered as the potential POIs that users served by the base station may visit. Based on this, more complex data processing is performed. After integrating research methods from several references and the prior research experience of the supervisor, we determined the research approach and wrote Python programs. The code developed can be used to identify activities for individual data points using the aforementioned data.

(Due to the ongoing paper writing process, the specific steps and code are not yet disclosed.)

2.1 Trajectory Similarity Measurement and Behavioral Activity Pattern Analysis

After generating users' activity chains (i.e., ATPs), a large amount of ATP data needs to be analyzed and processed. The approach involves measuring trajectory similarity and then performing clustering. For the clustering results, continuous adjustments are made to meet the needs of qualitative analysis, aiming to summarize the behavioral activity patterns of different groups.

2.2.1 Trajectory Similarity Measurement

Trajectory similarity measurement requires calculating the similarity matrix for all different ATPs. The current method used for similarity calculation is the Augmented Space-Time-Weighted Edit Distance, which is primarily based on the "multi-sequence alignment method based on dynamic programming" and optimized for processing time and spatial dimensional features. After obtaining the similarity matrix, spectral clustering is used to group all ATPs into clusters. Spectral clustering is a well-established clustering method. Based on the research needs and literature analysis, this study first divides the data into six categories:

1 Local Residents (age < 60)
2 Local Residents (age ≥ 60)
3 Travellers (age < 60)
4 Travellers (age ≥ 60)
5 Commuters (age < 60)
6 Commuters (age ≥ 60)

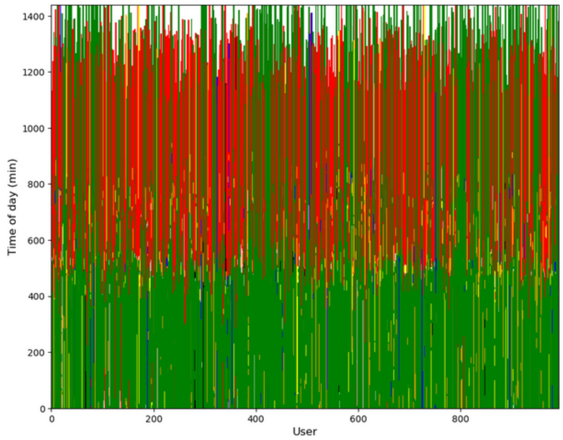
2.2.2 Individual Activity Pattern Analysis

Based on research needs and analysis from relevant literature, spectral clustering further divides each group into three categories. The results of the spectral clustering are shown in the figure below.

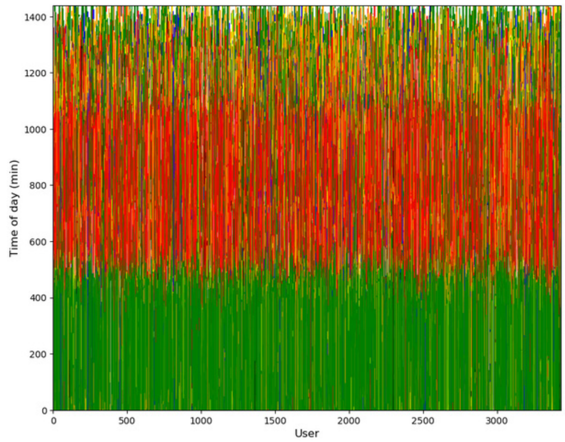
1 Local Residents, age < 60

This group consists of 7,821 users and 52,025 travel records. After excluding users with more than 8 activities, a total of 5,105 ATPs were input into the clustering algorithm.

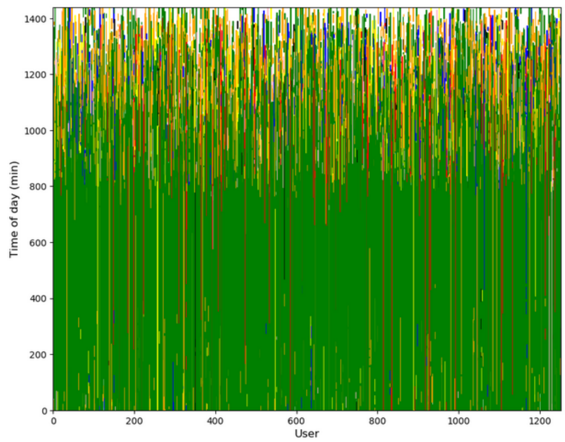
The visualization results of the clustering are shown in the figure below:



Group1 Cluster1
(Data Number: 602, Proportion: 11.8%)



Group1 Cluster2
(Data Number: 3421, Proportion: 67.0%)

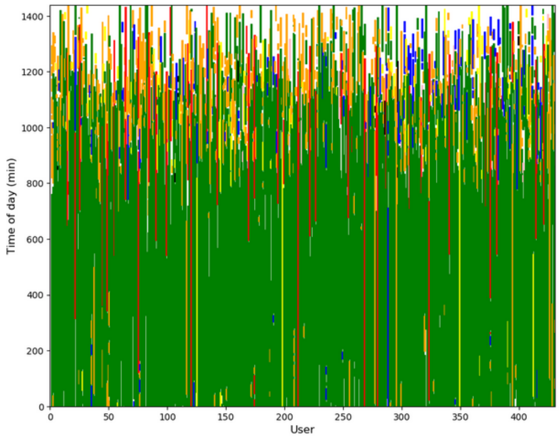


Group1 Cluster3
(Data Number: 1082, Proportion: 21.2%)

2 Local Residents, age ≥ 60

This group consists of 2,238 users and 15,930 travel records. After excluding users with more than 8 activities, a total of 1,411 ATPs were input into the clustering algorithm.

The visualization results of the clustering are shown in the figure below:



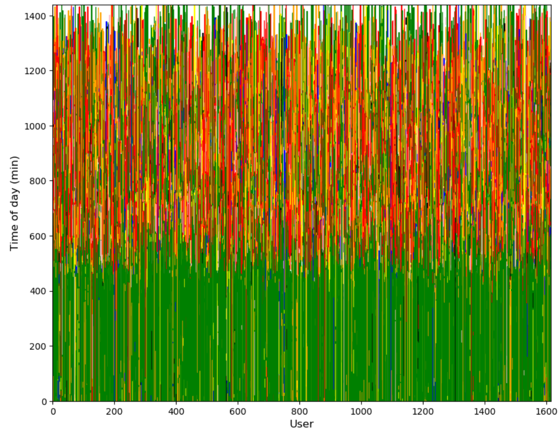
Group2 Cluster1
(Data Number: 307, Proportion: 26.2%)

Group2 Cluster2
(Data Number: 1041, Proportion: 73.8%)

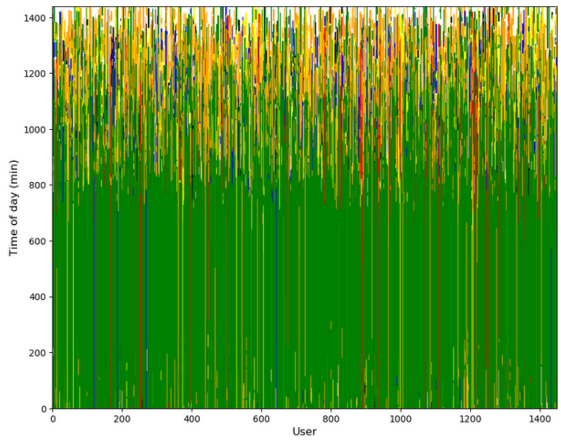
3 Travellers, age < 60

This group consists of 9,593 users and 74,783 travel records. After excluding users with more than 8 activities, a total of 5,855 ATPs were input into the clustering algorithm.

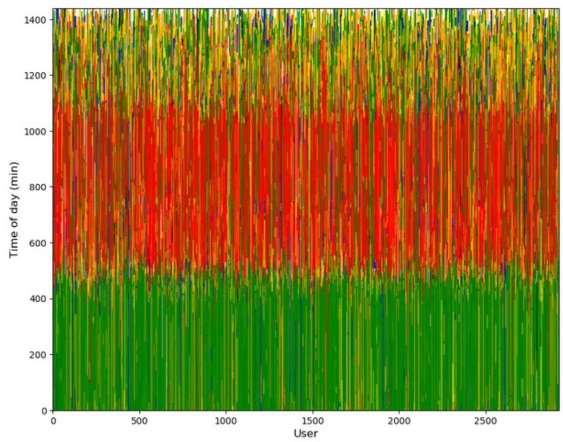
The visualization results of the clustering are shown in the figure below:



Group3 Cluster1
(Data Number: 1609, Proportion: 27.5%)



Group3 Cluster2
(Data Number: 1327, Proportion: 22.7%)



Group3 Cluster3
(Data Number: 2919, Proportion: 49.8%)

4 Travellers, age ≥ 60

This group consists of 2,792 users and 22,665 travel records. After excluding users with more than 8 activities, a total of 1,597 ATPs were input into the clustering algorithm.

The visualization results of the clustering are shown in the figure below:

Group4 Cluster1
(Data Number: 1161, Proportion: 72.3%)

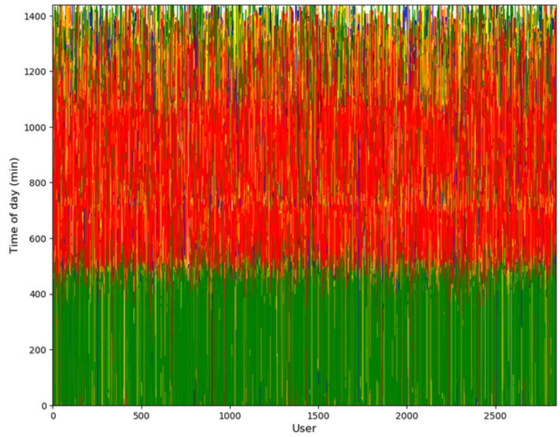
Group4 Cluster2
(Data Number: 436, Proportion: 27.7%)

Group1 Cluster3
(Data Number: 1082, Proportion: 21.2%)

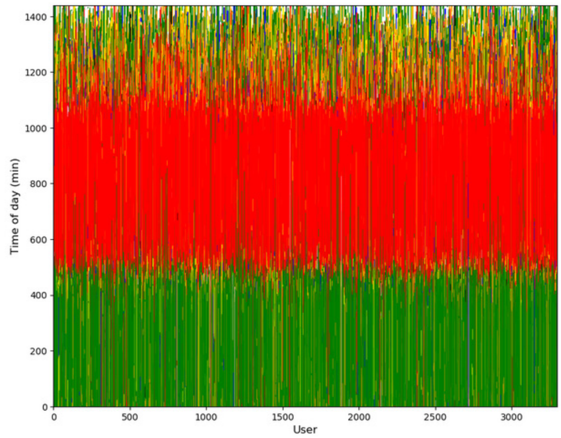
5 Commuters, age < 60

This group consists of 10,781 users and 76,461 travel records. After excluding users with more than 8 activities, a total of 6,887 ATPs were input into the clustering algorithm.

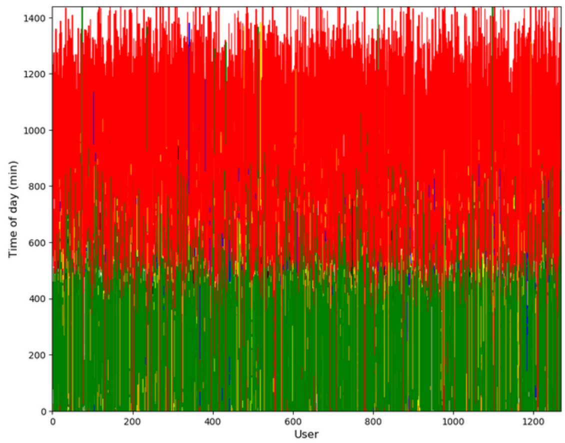
The visualization results of the clustering are shown in the figure below:



Group5 Cluster1
(Data Number: 2842, Proportion: 41.2%)



Group5 Cluster2
(Data Number: 3222, Proportion: 46.8%)

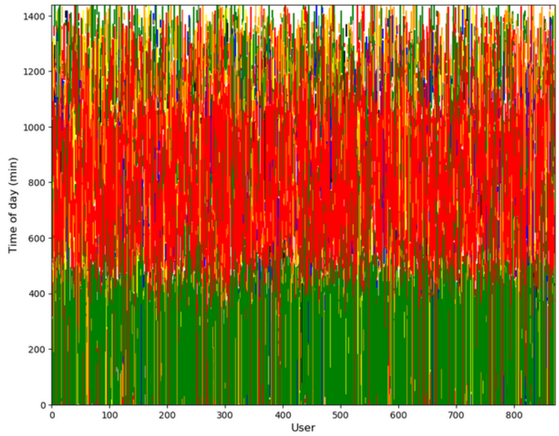


Group5 Cluster3
(Data Number: 823, Proportion: 12.0%)

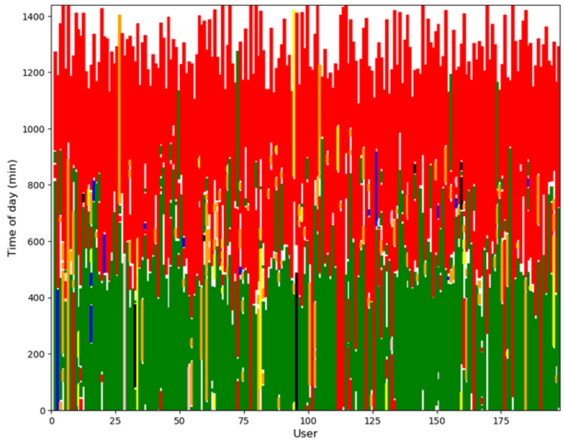
6 Commuters, age ≥ 60

This group consists of 1,776 users and 14,098 travel records. After excluding users with more than 8 activities, a total of 1,047 ATPs were input into the clustering algorithm.

The visualization results of the clustering are shown in the figure below:



Group6 Cluster1
(Data Number: 871, Proportion: 83.2%)



Group6 Cluster2
(Data Number: 176, Proportion: 16.4%)

Additional Explanation:

The research approach and steps of Phase 2 are rigorous and feasible. However, since the research is still ongoing, the final step of qualitative analysis is still in progress. The conclusion of the research is still in progress. The paper based on this research is currently in the writing stage, so some detailed steps of Phase 2 related to intellectual property have not been disclosed.

The paper based on this research is expected to be published before the start of the Fall semester of 2025.

Division of Work:

All of the content in Phase 1 and part of Phase 2 were independently completed by Jinyang Xu. Section 2.2.1 “Trajectory Similarity Measurement” in Phase 2 is the result of collaborative work by the team.

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